

DASC7011 Statistical Inference for Data Science

Chapter 3 Least Squares Estimation

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§3.1 Introduction

§3.2 Ordinary Least Squares Estimation (OLSE)

§3.3 Statistical Inference with OLSE

§3.4 Generalized Least Squares Estimation (GLSE)

A neural network model

§3.1 Introduction

- Least squares estimations are frequently used in neural network models.

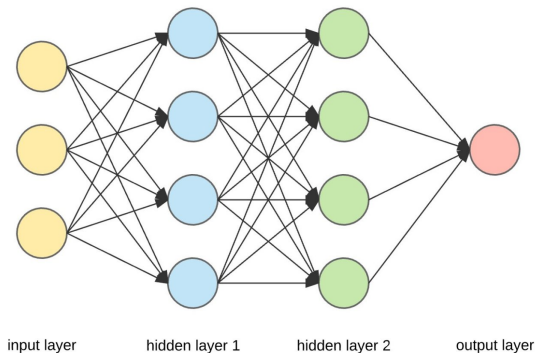


Figure 3.1: A neural network model.

Estimate the neural network model

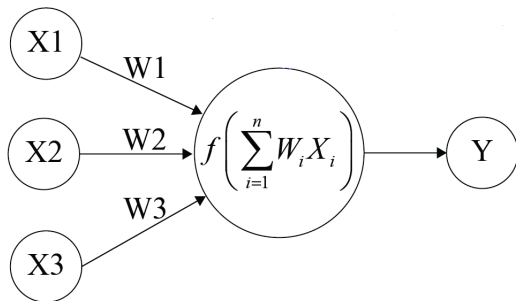


Figure 3.2: Estimating/Fitting the neural network model

- $f(\cdot)$ is called the *activation function*, usually chosen from several candidates based on the nature of Y .
- $\sum_{i=1}^n w_i X_i$ is a linear combination of X_i 's with weights w_i 's to be estimated.

Ideas of the estimation

- Expected (replace n with k):

$$\mathbb{E}[f^{-1}(Y)|\{X_i\}] = w_0 + w_1X_1 + \cdots + w_kX_k.$$

- Observed:

$$f^{-1}(Y_j) = w_0 + w_1X_{1j} + \cdots + w_kX_{kj} + e_j, \quad 1 \leq j \leq n,$$

where e_j 's are errors or deviations (from expectations).

- The smaller the errors, the better the model.
- Rule(s) are needed for fitting, prediction, and/or interpretation.

Ideas of the estimation

- Errors/Deviations:

$$e_j = f^{-1}(Y_j) - w_0 - w_1 X_{1j} - \cdots - w_k X_{kj}, \quad 1 \leq j \leq n.$$

- Possible rules:

(1) Minimize the sum of absolute errors $\sum_{i=1}^n |e_i|$.

(2) Minimize the sum of squared errors $\sum_{i=1}^n e_i^2$.

(3) Minimize the sum of 4th order errors $\sum_{i=1}^n e_i^4$.

(4)

- Among all, rule (2) is the most frequently used.
- The rule “minimizing the sum of squared errors” is usually referred to as **Least-Squares (LS)**, which was initially developed to estimate *regression models*.

Least Squares Estimation (LSE)

- A (general) regression model:

$$Y_i = f(X_{1i}, X_{2i}, \dots, X_{pi}; \boldsymbol{\theta}) + \varepsilon_i, \quad 1 \leq i \leq n, \quad (3.1)$$

where

- Y is the **response variable**,
- $\{X_k : 1 \leq k \leq p\}$ are **independent variables** or **regressors**,
- $f(\cdot)$ is a pre-specified function, or one from a known class of functions,
- $\boldsymbol{\theta}$ is (are) the parameter(s),
- $\{\varepsilon_i : 1 \leq i \leq n\}$ are random errors, usually assumed to be i.i.d. and follow certain distribution F *with mean 0*.

Least Squares Estimation (LSE)

- Suppose $f(\cdot)$ is fixed, and there is no unknown features or structures apart from parameter(s) θ .
- The LS algorithm estimates θ with the value(s) that minimize the following **Error Sum of Squares (ESS or SSE)** function,

$$S(\theta; f) = \sum_{i=1}^n [Y_i - f(X_{1i}, X_{2i}, \dots, X_{pi}; \theta)]^2. \quad (3.2)$$

We include f in the notation to indicate it depends on f too.

- In other words, the **least squares estimator (LSE)** of θ is

$$\hat{\theta}_f = \arg \min_{\theta} \left\{ S(\theta; f) \right\}. \quad (3.3)$$

- If f is allowed to be selected from a class $\mathcal{C}$, it could be selected/tuned as follows.

$$\hat{f} = \arg \min \left\{ S(\hat{\theta}_f; f) : f \in \mathcal{C} \right\}. \quad (3.4)$$

Ordinary Least Squares Estimation (OLSE)

§3.2 Ordinary Least Squares Estimation

3.2.1. Simple linear regression models.

- Let $(\mathbf{X}, \mathbf{Y}) = \{(X_i, Y_i) : 1 \leq i \leq n\}$ be a random sample from a **simple linear regression model**:

$$Y_i = \beta_0 + \beta_1 X_i + \varepsilon_i, \quad 1 \leq i \leq n, \quad (3.5)$$

where $\{\varepsilon_i\} \stackrel{i.i.d.}{\sim} F(0, \sigma^2)$. Parameters are $\boldsymbol{\theta} = (\beta_0, \beta_1)'$ and σ^2 .

- The *ordinary least squares* (OLS) algorithm estimates $\boldsymbol{\theta}$ by

$$\hat{\boldsymbol{\theta}} = \arg \min_{\boldsymbol{\theta}} \left\{ S(\beta_0, \beta_1) = \sum_{i=1}^n (Y_i - \beta_0 - \beta_1 X_i)^2 \right\}. \quad (3.6)$$

Ordinary Least Squares Estimation (OLSE)

- Solving (3.6),

$$\begin{cases} \frac{\partial S}{\partial \beta_0} = -2 \sum_{i=1}^n (Y_i - \beta_0 - \beta_1 X_i) = 0, \\ \frac{\partial S}{\partial \beta_1} = -2 \sum_{i=1}^n (Y_i - \beta_0 - \beta_1 X_i) X_i = 0, \end{cases}$$

gives

$$\hat{\beta}_1 = \frac{\sum_{i=1}^n (X_i - \bar{X})(Y_i - \bar{Y})}{\sum_{i=1}^n (X_i - \bar{X})^2}, \quad \hat{\beta}_0 = \bar{Y} - \hat{\beta}_1 \bar{X}. \quad (3.7)$$

- The OLS estimate of σ^2 is (chosen to be)

$$\hat{\sigma}^2 = \frac{1}{n-2} \sum_{i=1}^n (Y_i - \hat{\beta}_0 - \hat{\beta}_1 X_i)^2. \quad (3.8)$$

Example 3.1: simple LM

- **Example 3.1:** (*Simple linear regression model*) Read the data set saved in file `Stories.txt` into the R program by

```
mydata <- read.table('C:/Teaching/.../Stories.txt',  
                    header = TRUE)
```

- Suppose we want to regress the heights of buildings (variable `HGHT`) against the number of stories (variable `STORIES`) using a simple linear regression model $HGHT = \beta_0 + \beta_1 STORIES + \varepsilon$. We may do some preparing work to simplify coding:

```
# Preparing work  
View(mydata)  
Y <- mydata[,2]    # variable HGHT  
X <- mydata[,3]    # variable STORIES  
n <- nrow(mydata)  # sample size
```

Example 3.1: simple LM

- Find the OLS estimates using formulae (3.7) and (3.8):

```
> # Find OLS estimates by formulae
> y <- Y-mean(Y)    # deviation form of Y
> x <- X-mean(X)    # deviation form of X
> beta1 <- sum(y*x)/sum(x^2)
> beta0 <- mean(Y)-beta1*mean(X)
> sigma <- sqrt(sum((Y-beta0-beta1*X)^2)/(n-2))
> beta0; beta1; sigma
[1] 90.3096
[1] 11.29237
[1] 58.32593
```

- We can alternatively fit the model using the built-in function/algorithm `lm()`:

```
# Fit the model by lm()
fit <- lm(Y~X)    # Fit the model
summary(fit)      # Summary of the fitted model
```

Example 3.1: simple LM

- The summary of the fitted model:

```
Call:
lm(formula = Y ~ X)

Residuals:
    Min       1Q   Median       3Q      Max
-156.759  -33.239    5.995   28.450  167.487

Coefficients:
            Estimate Std. Error t value Pr(>|t|)
(Intercept)  90.3096   20.9622   4.308 6.44e-05 ***
X           11.2924    0.4844  23.310 < 2e-16 ***
---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

Residual standard error: 58.33 on 58 degrees of freedom
Multiple R-squared:  0.9036,    Adjusted R-squared:  0.9019
F-statistic: 543.4 on 1 and 58 DF,  p-value: < 2.2e-16
```



Multiple linear regression models

3.2.2. Multiple linear regression models.

- If there are more than one regressors, then we have a **multiple linear regression model**.
- A multiple linear regression model can be defined as

$$Y = \beta_0 + \beta_1 X_1 + \cdots + \beta_k X_k + \varepsilon, \quad (3.9)$$

where $\beta_0, \beta_1, \dots, \beta_k$ are unknown parameters, Y is the dependent variable, and $X_1, X_2, \dots, X_k$ are k independent variables/predictors/regressors.

- For observations $(Y_i, X_{1i}, X_{2i}, \dots, X_{ki})_{i=1,2,\dots,n}$, the model becomes

$$\begin{aligned} Y_i &= \beta_0 + \beta_1 X_{1i} + \beta_2 X_{2i} \cdots + \beta_k X_{ki} + \varepsilon_i, \\ i &= 1, \dots, n. \end{aligned} \quad (3.10)$$

Multiple linear regression models

- The multiple linear regression model (3.10) can be re-written into the following matrix form:

$$\begin{pmatrix} Y_1 \\ Y_2 \\ \vdots \\ Y_n \end{pmatrix} = \begin{pmatrix} 1 & X_{11} & \cdots & X_{k1} \\ 1 & X_{12} & \cdots & X_{k2} \\ \vdots & \vdots & \ddots & \vdots \\ 1 & X_{1n} & \cdots & X_{kn} \end{pmatrix} \begin{pmatrix} \beta_0 \\ \beta_1 \\ \vdots \\ \beta_k \end{pmatrix} + \begin{pmatrix} \varepsilon_1 \\ \varepsilon_2 \\ \vdots \\ \varepsilon_n \end{pmatrix}.$$

For convenience, we write

$$\mathbf{Y} = \mathbf{X}\boldsymbol{\beta} + \boldsymbol{\varepsilon}. \quad (3.11)$$

- We call matrix $\mathbf{X}$ the **design matrix**.
- Generally, we may assume that $k \ll n$, or at least, $k < n$.

Basic assumptions for multiple LM

- (A1) The relationship between $\mathbf{Y}$ and $\mathbf{X}$ is *linear* and is given by Eq. (3.11).
- (A2) The design matrix $\mathbf{X}$ is *non-stochastic*. In addition, no exact linear relationship exists between the independent variables, and hence, $\mathbf{X}$ is *fully ranked* ($\text{Rank}(\mathbf{X}) = k + 1 \leq n$).
- (A3) The error term has *zero mean* for all observations.
- (A4) The error term has *constant variance* σ^2 for all observations.
- (A5) Errors corresponding to different observations are mutually *independent* and therefore uncorrelated.
- (A6) The error term is *normally distributed*.

OLSE for multiple LM

- The **sum of squared errors/residuals (SSE)** function is defined as

$$\begin{aligned} S(\boldsymbol{\beta}) &= \sum_{i=1}^n (Y_i - \beta_0 - \beta_1 X_{1i} - \cdots - \beta_k X_{ki})^2 \\ &= (\mathbf{Y} - \mathbf{X}\boldsymbol{\beta})'(\mathbf{Y} - \mathbf{X}\boldsymbol{\beta}) \\ &= \mathbf{Y}'\mathbf{Y} - \boldsymbol{\beta}'\mathbf{X}'\mathbf{Y} - \mathbf{Y}'\mathbf{X}\boldsymbol{\beta} + \boldsymbol{\beta}'\mathbf{X}'\mathbf{X}\boldsymbol{\beta} \\ &= \boldsymbol{\beta}'\mathbf{X}'\mathbf{X}\boldsymbol{\beta} - 2\boldsymbol{\beta}'\mathbf{X}'\mathbf{Y} + \mathbf{Y}'\mathbf{Y}. \end{aligned}$$

- Differentiate is with respect to $\boldsymbol{\beta}$,

$$S'(\boldsymbol{\beta}) = 2\mathbf{X}'\mathbf{X}\boldsymbol{\beta} - 2\mathbf{X}'\mathbf{Y}.$$

- Equating it to zero, we obtain the OLSE of $\boldsymbol{\beta}$,

$$\hat{\boldsymbol{\beta}} = (\mathbf{X}'\mathbf{X})^{-1}\mathbf{X}'\mathbf{Y}. \quad (3.12)$$

Properties of OLS

- Under assumptions (A1) through (A5), $\hat{\beta}$ is **unbiased** since

$$\mathbb{E}(\hat{\beta}) = (\mathbf{X}'\mathbf{X})^{-1}\mathbf{X}'\mathbb{E}(\mathbf{Y}) = (\mathbf{X}'\mathbf{X})^{-1}\mathbf{X}'\mathbf{X}\beta = \beta.$$

- Under assumptions (A1) through (A5),

$$\begin{aligned}\text{Var}(\hat{\beta}) &= (\mathbf{X}'\mathbf{X})^{-1}\mathbf{X}'\text{Var}(\mathbf{Y})\mathbf{X}(\mathbf{X}'\mathbf{X})^{-1} \\ &= (\mathbf{X}'\mathbf{X})^{-1}\mathbf{X}'\sigma^2\mathbf{I}_n\mathbf{X}(\mathbf{X}'\mathbf{X})^{-1} \\ &= \sigma^2(\mathbf{X}'\mathbf{X})^{-1}.\end{aligned}$$

- Under assumptions (A1) through (A6), $\hat{\beta}$ is normal,

$$\hat{\beta} \sim N[\beta, \sigma^2(\mathbf{X}'\mathbf{X})^{-1}]. \quad (3.13)$$

- Under assumptions (A1) through (A5), $\hat{\beta}$ is *asymptotically normal*.

Estimate of error variance

- The population variance, or the error variance, σ^2 , is estimated by the **mean square error**:

$$\hat{\sigma}^2 = s^2 = \frac{\sum e_i^2}{n - k - 1}, \quad (3.14)$$

where $e_i = Y_i - \hat{Y}_i$ is the i th regression residual.

- The mean square error s^2 is an *unbiased* estimator of the error variance σ^2 , which has $(n - k - 1)$ degrees of freedom.
- The non-negative square-root of the residual variance s^2 , denoted by s , and sometimes SER, is called the **standard error of the regression**.

Gauss-Markov Theorem

Gauss-Markov Theorem: *Under assumptions (A1) through (A5), the OLS estimators $\hat{\beta}$ defined by Eq. (3.12) are the **best linear unbiased estimators (BLUE)** of β .*

Inferences on a single parameter

§3.3 Statistical inferences with OLSE

3.3.1. Inferences on a single parameter.

- Denote the OLSE of a single parameter β (we drop the index i for simplicity) by $\hat{\beta}$, and the estimated standard error (e.s.e) by s_β .
- By Property (3.13), the credible interval of β at the confidence level $(1 - \alpha)$ is

$$(\hat{\beta} - s_\beta \cdot t_{\alpha/2}(n - k - 1), \hat{\beta} + s_\beta \cdot t_{\alpha/2}(n - k - 1)), \quad (3.15)$$

where $t_{\alpha/2}(n - k - 1)$ is the $\frac{\alpha}{2}$ -th upper quantile of the $t(n - k - 1)$ distribution.

- The hypothesis $H_0 : \beta = b$ can be tested using the following T -test (statistic),

$$T = \frac{\hat{\beta} - b}{s_\beta} \sim t(n - k - 1). \quad (3.16)$$

Coefficient of determination

3.3.2. Inferences on multiple parameter.

- Inferences involving in multiple parameters needs more properties of the LSE.
- Define sums of squares (squared deviations):

$$\text{SST} = \sum (Y_i - \bar{Y})^2 = \mathbf{Y}'\mathbf{Y} - n\bar{Y}^2,$$

$$\text{SSR} = \sum (\hat{Y}_i - \bar{Y})^2 = \hat{\mathbf{Y}}'\hat{\mathbf{Y}} - n\bar{Y}^2,$$

$$\text{SSE} = \sum (Y_i - \hat{Y}_i)^2 = \sum e_i^2 = \mathbf{e}'\mathbf{e}.$$

- It is not difficult to see that

$$\begin{aligned}\text{SST} &= \sum (Y_i - \bar{Y})^2 = \sum (Y_i - \hat{Y}_i + \hat{Y}_i - \bar{Y})^2 \\ &= \sum (\hat{Y}_i - \bar{Y})^2 + \sum e_i^2 + 2 \sum e_i(\hat{Y}_i - \bar{Y}) \\ &= \text{SSR} + \text{SSE}.\end{aligned}$$

Coefficient of determination

- The last equation holds because
 - The residual vector $\mathbf{e}$ is orthogonal to each column of $\mathbf{X}$, including the constant column/vector $\mathbf{1} = (1, \dots, 1)'$, since

$$\begin{aligned}\mathbf{X}'\mathbf{e} &= \mathbf{X}'(\mathbf{Y} - \hat{\mathbf{Y}}) \\ &= \mathbf{X}'[\mathbf{I} - \mathbf{X}(\mathbf{X}'\mathbf{X})^{-1}\mathbf{X}']\mathbf{Y} \\ &= [\mathbf{X}' - \mathbf{X}'\mathbf{X}(\mathbf{X}'\mathbf{X})^{-1}\mathbf{X}']\mathbf{Y} \\ &= (\mathbf{X}' - \mathbf{X}')\mathbf{Y} = \mathbf{0}.\end{aligned}$$

- Denote $\mathbf{R} = \mathbf{X}(\mathbf{X}'\mathbf{X})^{-1}\mathbf{X}'$, then $\hat{\mathbf{Y}} = \mathbf{R}\mathbf{Y}$. The residual vector $\mathbf{e}$ is orthogonal to the fitted response vector $\hat{\mathbf{Y}}$ since $\mathbf{R}' = \mathbf{R}$, $\mathbf{R} = \mathbf{R}^2$, and

$$\hat{\mathbf{Y}}'\mathbf{e} = \mathbf{Y}'\mathbf{R}'(\mathbf{I} - \mathbf{R})\mathbf{Y} = \mathbf{Y}'\mathbf{R}\mathbf{Y} - \mathbf{Y}'\mathbf{R}^2\mathbf{Y} = \mathbf{0}.$$

Coefficient of determination

- The sums of squares are measures of *total variation* of Y , variations *explained* by the model, and variations *not explained* by the model (of residuals), respectively.
- The **coefficient of determination** is defined as the ratio of the sum of squares due to regression to the total sum of squares:

$$R^2 = \frac{SSR}{SST} = 1 - \frac{SSE}{SST}. \quad (3.17)$$

- By definition, R^2 is a measure (in percentage) of how much the total variation of Y is accounted for, or explained, by the model. Or equivalently, how well the independent variables can explain Y in terms of its variation.
- Roughly speaking, the larger is R^2 , the better is the fitted model.

Adjusted coefficient of determination

- The **adjusted R -square**, denoted as R_a^2 , considers the ratio of means of squares:

$$R_a^2 = 1 - \frac{S_e^2}{S_Y^2} = 1 - \frac{\text{SSE}/(n - k - 1)}{\text{SST}/(n - 1)}. \quad (3.18)$$

- Notice that

$$\begin{aligned} 1 - R_a^2 &= \frac{\text{SSE}/(n - k - 1)}{\text{SST}/(n - 1)} = \frac{\text{SSE}}{\text{SST}} \cdot \frac{n - 1}{n - k - 1} \\ &> \frac{\text{SSE}}{\text{SST}} = 1 - R^2. \end{aligned}$$

Therefore,

$$0 \leq R_a^2 < R^2 \leq 1.$$

- R^2 and R_a^2 are measures of the **goodness-of-fit (GOF)** of the model.

Inferences on multiple parameters

- Suppose that we want to test the following null hypothesis

$$H_0 : \beta_k = \beta_{k-1} = \cdots = \beta_{k-p+1} = 0$$

against alternative

$$H_1 : \text{not all of } \beta_k, \beta_{k-1}, \cdots, \beta_{k-p+1} \text{ are zeros,}$$

where $1 \leq p \leq k$ are two integers.

- Under the null hypothesis, the model becomes

$$Y = \beta_0 + \beta_1 X_1 + \cdots + \beta_{k-p} X_{k-p} + \varepsilon. \quad (3.19)$$

- We call model (3.19) the **restricted model**, in contrast to the **unrestricted model** or **full model** defined by Eq. (3.9).

Inferences on multiple parameters

- Here an F -test is applied. The test statistic is

$$F = \frac{(\text{SSE}_R - \text{SSE}_F)/p}{\text{SSE}_F/(n - k - 1)} \sim F(p, n - k - 1) \quad \text{under } H_0, \quad (3.20)$$

where SSE_R and SSE_F are the sum of squared residuals functions for the restricted model and the full model, respectively.

- For a given significance level $\alpha > 0$, the critical region is $F > F_\alpha(p, n - k - 1)$, the α -th upper quantile of the $F(p, n - k - 1)$ distribution.
- This F test applies for testing hypothesis involving a single parameter too, and is equivalent to the t test.

A goodness-of-fit test

- Another special example of the F test is the case that $p = k$. In this case, the restricted model becomes a *null model*

$$Y = \beta_0 + \varepsilon.$$

- In other words, the null hypothesis suggests that all predictors in model (3.9) are superfluous.
- If the null hypothesis is rejected, then we say that the regression model (3.9) is *significant* (significantly better than the null model). Otherwise, the model is *insignificant* or *trivial*.
- Such an F test is usually referred to as the *goodness-of-fit test*.

3.3.3. Predictions.

- Consider the multiple regression model (3.9) with OLSE (and BLUE) $\hat{\boldsymbol{\beta}} = (\hat{\beta}_0, \hat{\beta}_1, \dots, \hat{\beta}_k)'$.
- Given a special observation of predictors $(X_{10}, X_{20}, \dots, X_{k0})$, the corresponding $\mathbb{E}(Y_0)$ is predicted by

$$\hat{Y}_0 = \hat{\beta}_0 + \hat{\beta}_1 X_{10} + \dots + \hat{\beta}_k X_{k0}. \quad (3.21)$$

- If we denote the new observation as a (column) vector $\mathbf{X}_0 = (1, X_{10}, \dots, X_{k0})'$, then

$$\hat{Y}_0 = \mathbf{X}_0' \hat{\boldsymbol{\beta}}.$$

- The prediction $\hat{Y}_0$ is actually an estimate of the *conditional expected value* of Y given $(X_1, \dots, X_k) = (X_{10}, \dots, X_{k0})$, i.e., $\mathbb{E}(Y|\mathbf{X}_0)$. We use the notation $\mathbb{E}(Y_0)$ for simplicity.

Prediction or Forecast

- The prediction $\hat{Y}_0$ is a linear combination of the BLUE $\hat{\beta}$, and thus a BLUE of $\mathbb{E}(Y_0)$.
- Moreover, $\hat{Y}_0$ is normally distributed with variance

$$\begin{aligned}\sigma_{Y_0}^2 &= \text{Var}(\mathbf{X}'_0 \hat{\beta}) = \text{Var}[\mathbf{X}'_0 (\mathbf{X}'\mathbf{X})^{-1} \mathbf{X}'\mathbf{Y}] \\ &= \mathbf{X}'_0 (\mathbf{X}'\mathbf{X})^{-1} \mathbf{X}' \text{Var}(\mathbf{Y}) \mathbf{X} (\mathbf{X}'\mathbf{X})^{-1} \mathbf{X}_0 \\ &= \mathbf{X}'_0 (\mathbf{X}'\mathbf{X})^{-1} \mathbf{X}' \sigma^2 \mathbf{I}_n \mathbf{X} (\mathbf{X}'\mathbf{X})^{-1} \mathbf{X}_0 \\ &= \sigma^2 \mathbf{X}'_0 (\mathbf{X}'\mathbf{X})^{-1} \mathbf{X}_0.\end{aligned}$$

- The estimated value of this variance is

$$s_{Y_0}^2 = s^2 \mathbf{X}'_0 (\mathbf{X}'\mathbf{X})^{-1} \mathbf{X}_0,$$

where s^2 is the estimated residual variance.

Confidence interval

- Therefore, the **confidence interval** at the γ -level of confidence (e.g., $\gamma = 0.95$) for $\mathbb{E}(Y_0)$ is

$$CI_\gamma = \hat{Y}_0 \pm t_{(1-\gamma)/2}(n-k-1)s_{Y_0}. \quad (3.22)$$

- Interpretation:

$$\mathbb{P}\left(\mathbb{E}(Y) \in CI_\gamma \mid \mathbf{X}, \mathbf{X}_0\right) = \gamma.$$

- The **prediction/forecast error** of $\hat{Y}_0$ is

$$\begin{aligned}e_0 &= Y_0 - \hat{Y}_0 \\&= \mathbf{X}'_0 \boldsymbol{\beta} + \varepsilon_0 - \mathbf{X}'_0 \hat{\boldsymbol{\beta}} \\&= \mathbf{X}'_0 (\boldsymbol{\beta} - \hat{\boldsymbol{\beta}}) + \varepsilon_0.\end{aligned}$$

- The variance of the prediction error is

$$\begin{aligned}\sigma_{F_0}^2 &= \text{Var}[\mathbf{X}'_0 (\boldsymbol{\beta} - \hat{\boldsymbol{\beta}})] + \text{Var}(\varepsilon_0) \\&= \mathbf{X}'_0 (\mathbf{X}'\mathbf{X})^{-1} \sigma^2 \mathbf{X}_0 + \sigma^2 \\&= \sigma^2 [1 + \mathbf{X}'_0 (\mathbf{X}'\mathbf{X})^{-1} \mathbf{X}_0].\end{aligned}$$

Prediction interval

- The estimated variance of the prediction error is

$$s_{F_0}^2 = s^2[1 + \mathbf{X}_0'(\mathbf{X}'\mathbf{X})^{-1}\mathbf{X}_0].$$

- Therefore, the **prediction interval** at the γ -level of confidence is

$$PI_\gamma = \hat{Y}_0 \pm t_{(1-\gamma)/2}(n-k-1)s_{F_0}. \quad (3.23)$$

- Understanding:

$$\mathbb{P}(Y \in PI_\gamma | \mathbf{X}, \mathbf{X}_0) = \gamma.$$

- Since $s_{F_0}^2 > s_{Y_0}^2$, the prediction interval is always *wider* than the corresponding (at the same confidence level) confidence interval.

Example 3.2: Crime Data

- **Example 3.2:** Analyze the Freedman data from R package car (Companion to Applied Regression).
- Load data: `library(car); data(Freedman); Freedman`.
- Observations from 110 U.S. metropolitan areas with 1968 populations of 250,000 or more, with some missing data.
- Four variables:
 - **population:** Total 1968 population, in thousands.
 - **nonwhite:** Percent nonwhite population, 1960.
 - **density:** Population per square mile, 1968.
 - **crime:** Crime rate per 100,000, 1969.

Example 3.2: Crime Data

- **Purpose:** investigate effects of other three variables on the crime rate.
- First model named as `fit0` in R:
 - R coding:

```
fit0 <- lm(crime ~ ., data = Freedman)
summary(fit0)
```

- Summary of the fit, `summary(fit0)`:

Coefficients	Estimate	Std.Error	t-value	Pr(> t)
(Intercept)	2193.70088	143.04566	15.336	< 2e-16
population	0.24495	0.06095	4.019	0.000116
nonwhite	26.03770	8.76746	2.970	0.003764
density	-0.02145	0.06578	-0.326	0.745045

Example 3.2: Crime Data

- Effect of **density** is insignificant by the t -test. Remove it from the model.

- R coding:

```
fit1 <- lm(crime ~ population + nonwhite,  
           data = Freedman)
```

- Summary:

(Intercept)	2.185e+03	1.398e+02	15.630	< 2e-16
population	2.376e-01	5.639e-02	4.214	5.63e-05
nonwhite	2.611e+01	8.724e+00	2.993	0.0035

Multiple R-squared: 0.2279 (R^2)

Adjusted R-squared: 0.212 (R_a^2)

F-statistic: **14.32** on 2 and 97 DF, p-value: 3.56e-06

Example 3.2: Crime Data

- All coefficients are significant at the 5% level.
- The summary also provides the overall GOF test results:
 $F = 14.32$, null distribution $F(2, 97)$, p -value $3.56e-06$.
- To find SST, SSR and SSE, we look at the ANOVA table produced by `avona(fit1)`:

	Df	Sum Sq	Mean Sq	F value	Pr(>F)
population	1	15000803	15000803	19.678	2.421e-05
nonwhite	1	6828890	6828890	8.958	0.003504
Residuals	97	73945387	762324		

- It is not difficult to check that

$$F = \frac{(15000803 + 6828890)/2}{73945387/97} = 14.31787.$$

- Conclusion: the model is significant.

Example 3.2: Crime Data

- To conduct the F -test for `fit1` against `fit0`, we need the ANOVA table of `fit0`.

- ANOVA of `fit0`:

	Df	Sum Sq	Mean Sq	F value	Pr(>F)
population	1	15000803	15000803	19.678	2.421e-05
nonwhite	1	6828890	6828890	8.958	0.003504
density	1	81829	81829	0.1064	0.74504
Residuals	96	73863558	769412		

- F -test by manual calculation:

$$F = \frac{(73945387 - 73863558)/1}{73863558/96} = \frac{81829/1}{73863558/96} = 0.1063526.$$

- The p -value is `pf(F.stat, 1, 96, low.tail=F)` = 0.7450453.
We cannot reject `fit1` in favor of `fit0`.

Example 3.2: Crime Data

- The predictions can be calculated by either of the following R commands:
 - Command “`fit1$fitted.values`” displays all predictions for the original data set (with non-omitted explanatory values). These predictions are usually called *fitted values*.
 - Command “`predict.lm(fit1, newdata = Freedman)`” calculates predictions for any *new data*.
- Command “`predict.lm(fit1, Freedman, interval = "confidence", level = 0.95)`” calculates predictions as well as confidence intervals.
- Command “`predict.lm(fit1, Freedman, interval = "prediction", level = 0.95)`” calculates predictions as well as prediction intervals. □

Generalized Least Squares Estimation (GLSE)

§3.4 Generalized Least Squares Estimation¹

- Good properties of OLSEs, especially the (asymptotical) normality and consistency, make statistical inferences relatively simple.
- These good properties are ensured by the basic assumptions, especially the following three.

(A2) Explanators are not random.

(A4) Errors $\{X_i\}$ have constant variance (homoscedastic).

(A5) Errors $\{X_i\}$ are serially uncorrelated.

- However, in practice, these assumptions are frequently disrupted. Consequently, OLSEs could be inefficient, inconsistent, and/or non-normal.

- **Generalized Least Squares Estimation (GLSE)** is needed.

¹More details are introduced in STAT8007: *Statistical methods in economics and finance*.

Heteroscedasticity

3.4.1: GLSE for LM with heteroscedastic errors

- For illustration, consider the following rent-income data.

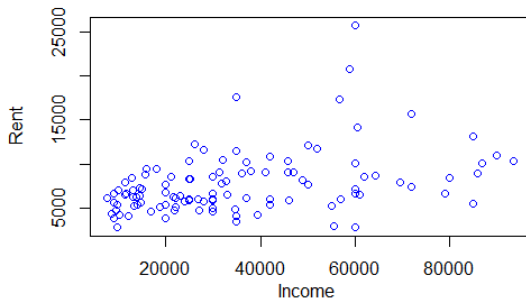


Figure 3.3: Annual rents and incomes for a sample of New Yorkers.

- We are estimating a simple LM for this,

$$Rent_i = \beta_0 + \beta_1 Income_i + \varepsilon_i.$$

Heteroscedasticity

- **Observation:** data with larger incomes are more diversified.
- **Check understanding:** Consumers with low values of income have little scope for varying their rent expenditures, and hence $\text{Var}(\varepsilon_i)$ is low. On the other hand, wealthy consumers can choose to spend a lot of money on rent, or to spend less, depending on tastes, as a result, $\text{Var}(\varepsilon_i)$ is high.
- **Heteroscedasticity** presents in the model: the variance of ε_i , and hence of Rent_i , is NOT a constant σ^2 .
- In general, we have

$$\text{Var}(\varepsilon_i) = \sigma_i^2, \quad i = 1, 2, \dots, n.$$

- In other words, basic assumption (A4) is destroyed.

Problems with heteroscedasticity

- Under heteroscedasticity, OLSEs are still **unbiased** and **consistent**.
- However, OLSEs are **inefficient** — they are no longer the BLUEs.
- More importantly, SER in formulas (3.8) or (3.14) are **WRONG!**
There is NOT any unified σ^2 to be estimated.
- Consequently,
 - T -tests based on OLS SER are **WRONG**.
 - F -tests (need homoscedasticity for the F -distribution) are **WRONG**.
 - Confidence intervals and prediction intervals are **invalid**.

GLSE for LM with heteroscedastic errors

- For simplicity, rewrite the model as

$$Y_i = \beta_0 + \beta_1 X_i + \varepsilon_i, \quad \text{Var}(\varepsilon_i) = \sigma_i^2, \quad 1 \leq i \leq n.$$

- Basic assumptions other than (A4) are satisfied.
- Taking into account our comprehension of the data's characteristics, we may *assume* that error variances are proportional to incomes, that is,

$$\sigma_i^2 = \sigma^2 X_i \quad \text{for some constant } \sigma^2 > 0.$$

- Based on this assumption, we may *transform* the data as follows.

$$\tilde{Y}_i = \frac{Y_i}{\sqrt{X_i}}, \quad \tilde{X}_{1i} = \frac{1}{\sqrt{X_i}}, \quad \tilde{X}_{2i} = \sqrt{X_i}, \quad \tilde{\varepsilon}_i = \frac{\varepsilon_i}{\sqrt{X_i}}.$$

GLSE for LM with heteroscedastic errors

- Then, the model can be rewritten into

$$\tilde{Y}_i = \beta_0 \tilde{X}_{1i} + \beta_1 \tilde{X}_{2i} + \tilde{\varepsilon}_i. \quad (3.24)$$

- Notice that the transformed model (3.24) is a multiple linear model without intercept (or, *through the origin*).
- It is not difficult to check that all basic assumptions (A1) through (A6) are satisfied.
- Finally, apply OLS to the transformed model (3.24).
- Statistical inferences introduced in §3.3 become valid.
- We call the above process of estimation the **Generalized Least Squares Estimation (GLSE)**.

Weighted Least Squares Estimation (WLSE)

- Notice that the SSE functions for OLSE (S_o) and GLSE (S_g) are

$$\begin{aligned} S_o(\beta_0, \beta_1) &= \sum_{i=1}^n (Y_i - \beta_0 - \beta_1 X_i)^2, \\ S_g(\beta_0, \beta_1) &= \sum_{i=1}^n \left(\tilde{Y}_i - \beta_0 \tilde{X}_{1i} - \beta_1 \tilde{X}_{2i} \right)^2, \\ &= \sum_{i=1}^n \frac{(Y_i - \beta_0 - \beta_1 X_i)^2}{\textcolor{red}{X}_i}. \end{aligned}$$

- Comparing these two SSEs, we see that a sequence of *weights* $\{W_i = X_i^{-1}\}$ are added to terms in the summation.
- Therefore, such a GLSE is usually referred to the **Weighted Least Squares Estimation (WLSE)**.

Feasible GLSE (FGLSE)

- The aforesaid GLSE is based on our assumption of the heteroscedasticity: $\sigma_i^2 \propto X_i$.
- We are not quite sure whether the transformed errors $\{\tilde{\varepsilon}_i\}$ are homoscedastic. The **Breusch-Pagan test**, including the **White's test** as a special case, can be used for this problem. The test should be applied to the original errors $\{\varepsilon_i\}$ before conducting the GLSE process.
- If we assume $\sigma_i^2 \propto X_i^2$, another sequence of weights, $\{W_i = X_i^{-2}\}$.
- We can make a weaker but more feasible assumption: $\sigma_i^2 \propto X_i^d$, where $d > 0$ is an unknown parameter to be estimated or tuned. In this case, we call this the **Feasible Generalized Least Squares Estimation (FGLSE)**.

Example 3.3: Rent-Income

- **Example 3.3:** Consider the rent-income data and suppose that we want to estimate the relationship

$$Y_i = \beta_0 + \beta_1 X_i + \varepsilon_i,$$

where Y_i and X_i stand for rent expenditure and income, respectively.

- The OLSEs of coefficients are summarized as follows.

Coefficients:

	Estimate	Std. Error	t value	Pr(> t)	
(Intercept)	5.455e+03	6.028e+02	9.051	7.67e-15	***
x	6.357e-02	1.439e-02	4.418	2.42e-05	***

Example 3.3: Rent-Income

- Heteroscedasticity is tested to be significant.

```
> library(lmtest)
> bptest(fit0)

studentized Breusch-Pagan test

data: fit0
BP = 5.607, df = 1, p-value = 0.01789
```

- The GLSE in Eq. (3.24) can be manually conducted using the following R codes.

```
# Manual GLSE: transform the data and OLS
x1 <- 1/sqrt(x)
x2 <- sqrt(x)
y1 <- y/sqrt(x)
fit1 <- lm(y1 ~ x1+x2-1) # LM without intercept
```

Example 3.3: Rent-Income

- The GLSEs of coefficients are given below.

```
> summary(fit1)$coefficients
```

	Estimate	Std. Error	t value	Pr(> t)
x1	5.085513e+03	411.42410002	12.360755	2.851614e-22
x2	7.396252e-02	0.01426932	5.183325	1.049165e-06

- Heteroscedasticity becomes insignificant in the transformed model.

```
> bptest(fit1)
```

studentized Breusch-Pagan test

data: fit1

BP = 1.7005, df = 1, p-value = 0.1922

Example 3.3: Rent-Income

- The GLSE/WLSE can be automatically conducted using the following codes.

```
# WLSE: Define weights and WLS  
w <- 1/x  
fit2 <- lm(y ~ x, weights = w)  
summary(fit2)$coefficients
```

- The same estimations, as by manual GLSE, are obtained.
- However, when we apply the BP test to `fit2`, the same testing results, as by manual OLSE, will be obtained.
- This is because we modified the SSE function but didn't do any transformation to the data. □

GLSE for LM with serial correlation

3.4.2: GLSE for LM with serial correlation

- **Serial correlation** occurs in a time series data $\{X_t\}$ when X_t is correlated with some lagged version of itself, e.g., X_{t-1} .
- When considering regression models with time series data such as annual GDPs, monthly inflations, etc., we have to be aware of possible serial correlation in the data.
- For example, we are regression monthly CPIs Y_t against monthly Inflations X_t ,

$$Y_t = \beta_0 + \beta_1 X_t + \varepsilon_t, \quad 1 \leq t \leq T.$$

Both $\{X_t\}$ and $\{Y_t\}$, and hence $\{\varepsilon_t\}$, are (common sense) serially correlated.

- OLSE for such an LM with serial correlation will be unbiased, consistent, but **inefficient**. More importantly, **estimated SER is invalid**.

GLSE for LM with serial correlation

- Define a null hypothesis:

$$H_0 : \{\varepsilon_t\} \text{ is serially uncorrelated.}$$

- The **Durbin-Watson test** can be used to test the *presence of lag 1 serial correlation*, i.e., to test H_0 against

$$H_a : \mathbb{E}(\varepsilon_t \varepsilon_{t-1}) \neq 0.$$

- The **Breusch-Godfrey test** tests H_0 against

$$H_a : \text{disturbances are serially correlated among the first } L \text{ lags (up to lag } L),$$

where $L \geq 1$ is a pre-specified upper bound of lags.

GLSE for LM with serial correlation

- When serial correlation presents, like for heteroscedasticity, we need an assumption on the structure of the serial correlation to conduct the GLSE.
- The most commonly used structure is: assume $\{\varepsilon_t\}$ are firstly order autoregressive,

$$\varepsilon_t = \rho\varepsilon_{t-1} + \tilde{\varepsilon}_t,$$

where $|\rho| < 1$ is an unknown constant, and $\{\tilde{\varepsilon}_t\}$ are i.i.d with mean 0 and variance σ^2 .

- *If ρ is known*, we may transform the data as follows,

$$\tilde{Y}_t = Y_t - \rho Y_{t-1}, \quad \tilde{X}_t = X_t - \rho X_{t-1}, \quad 2 \leq t \leq T.$$

- The transformed model becomes

$$\tilde{Y}_t = \tilde{\beta}_0 + \beta_1 \tilde{X}_t + \tilde{\varepsilon}_t, \quad 2 \leq t \leq T,$$

where $\tilde{\beta}_0 = \beta_0(1 - \rho)$.

GLSE for LM with serial correlation

- However, ρ is actually unknown. We may *pre-estimate* it using the OLS residuals.
- Such a GLSE is referred to as the **Cochrane-Orcutt** estimation.
- Another slightly different but more efficient estimation is the **Prais-Winsten** estimation. It rescues the first observation ($t = 1$) and uses certain iterative computation to improve the accuracy and efficiency of the estimation.
- A third method is the **Non-linear Least Squares Estimation (NLSE)**, the following non-linear regression model is estimated on the whole.

$$Y_t = \rho Y_{t-1} + \beta_0(1 - \rho) + \beta_1(X_t - X_{t-1}) + \tilde{\varepsilon}_t, \quad 1 \leq t \leq T.$$

Consistent estimated standard errors

3.4.3: Consistent estimated standard errors

- GLSEs are not automated.
- Assumptions on the structure of heteroscedasticity and/or serial correlation can be sometime difficult.
- Nevertheless, OLSEs are not that bad — they are unbiased and consistent.
- A *shortcut* frequently in econometrics is: *using OLSE and revising the formulas for estimated standard errors* to ensure valid statistical inferences.
- Notice that the OLSE of each coefficient β is a linear function of Y_i 's. Denote it as

$$\hat{\beta} = \sum_{i=1}^n w_i Y_i.$$

Consistent estimated standard errors

- The **White's consistent standard errors** for $\hat{\beta}$ is defined as

$$e.s.e.(\hat{\beta}) = \sqrt{\sum_{i=1}^n w_i^2 e_i^2},$$

where e_i 's are OLS residuals.

- For LM with serial correlation, or both heteroscedasticity and serial correlation, we use the the following **Newey-West consistent standard errors**,

$$e.s.e(\hat{\beta}) = \sqrt{\sum_{t=1}^T \sum_{s=t-L}^{t+L} w_t w_s e_t e_s},$$

where e_i 's are OLS residuals, and $L > 0$ is a pre-specified integer standing for the order of serial correlation.

Example 3.4: the rent-income model

- **Example 3.4:** Consistent estimated standard errors.
- Consider the rent-income data first.
- The White's consistent standard errors, as well as corresponding tests, are conducted using the following R codes.

```
# white's Robust Standard Errors for rent-income model
library(sandwich)
summary.white <- function(model) {
  print(coeftest(model, vcov. = vcovHC))
  print(waldtest(model, vcov = vcovHC))
}                                     # Define a function
summary(fit0)
summary.white(fit0)
```

- R string `vcovHC` stands for **H**eteroscedasticity **C**onsistent variances and covariances.

Example 3.4: the rent-income model

- Results from OLSE with White's consistent standard errors:

```
> summary.white(fit0)
```

t test of coefficients:

	Estimate	Std. Error	t value	Pr(> t)
(Intercept)	5.4555e+03	4.0984e+02	13.311	< 2.2e-16 ***
x	6.3568e-02	1.5060e-02	4.221	5.151e-05 ***

Wald test

Model 1: $y \sim x$
Model 2: $y \sim 1$

	Res.Df	Df	F	Pr(>F)
1	106			
2	107	-1	17.817	5.151e-05 ***

Example 3.4: the rent-income model

- Compared with results from pure OLSE:

```
> summary(fit0)
Coefficients:
              Estimate Std. Error t value Pr(>|t|)
(Intercept) 5.455e+03  6.028e+02  9.051 7.67e-15 ***
x           6.357e-02  1.439e-02  4.418 2.42e-05 ***
---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

Residual standard error: 3303 on 106 degrees of freedom
Multiple R-squared:  0.1555,    Adjusted R-squared:  0.1475
F-statistic: 19.51 on 1 and 106 DF,  p-value: 2.417e-05
```

- Estimates of coefficients are the same.
- Estimated standard errors and test results are revised.

Example 3.4: the poverty-unemployment model

- Consider one more data: poverty rate (variable P) and unemployment rate (variable U) for $T = 24$ years (from 1980 to 2003).
- The economists ask: *How much does the poverty rate rise when the unemployment rate rises?*
- The following simple regression model is used to address this question.

$$P_t = \beta_0 + \beta_1 U_t + \varepsilon_t, \quad 1 \leq t \leq T.$$

- This is a regression model with time series data.

Example 3.4: the poverty-unemployment model

- Both the **Durbin-Watson test** and the **Breusch-Godfrey test** show that serial correlations are significant.
- Both GLSEs, the **Cochrane-Orcutt Estimation** and the **Prais-Winsten Estimation**, are conducted.
- The NLSE is also provided in the R script.
- We focus on statistical inferences with the Newey-West consistent standard errors given by R string `vcovHAC`.
 - HAC stands for Heteroscedasticity and Autocorrelation Consistent.

Example 3.4: the poverty-unemployment model

- Results from OLSE and the Newey-West consistent SE.

```
> summary.nw(fit.ols)

t test of coefficients:

              Estimate Std. Error t value  Pr(>|t|)
(Intercept)  9.792052   0.638355  15.3395 3.128e-13 ***
U             0.586614   0.096963   6.0499 4.339e-06 ***
---
Signif. codes:
0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

wald test

Model 1: P ~ U
Model 2: P ~ 1
      Res.Df Df      F      Pr(>F)
1         22
2         23 -1 36.601 4.339e-06 ***
```

Example 3.4: the poverty-unemployment model

- Results from pure OLSE.

```
> summary(fit.ols)
```

Coefficients:

	Estimate	Std. Error	t value	Pr(> t)	
(Intercept)	9.79205	0.61119	16.021	1.30e-13	***
U	0.58661	0.09473	6.193	3.12e-06	***

Signif. codes:

0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

Residual standard error: 0.6763 on 22 degrees of freedom

Multiple R-squared: 0.6355, Adjusted R-squared: 0.6189

F-statistic: 38.35 on 1 and 22 DF, p-value: 3.116e-06

